

Tetracyclines and the risk of *Clostridioides difficile* infection

Evidence summary and proposal for a Bayesian meta-analysis

Document prepared as a medical-student graduation thesis proposal — Infectious Diseases

Part 1 — Evidence summary

The clinical question

The choice of antibiotic in patients at high risk of *Clostridioides difficile* infection (CDI) remains a practical dilemma. It has long been observed that tetracyclines — particularly doxycycline — may be associated with a **lower** risk of CDI than other antibiotic classes, but individual study results are partly conflicting.

What the data show

The evidence comes from three converging lines:

1. **Class-comparison meta-analyses.** In two independent community-associated CDI meta-analyses (2013), tetracyclines are the only class that does **not** increase risk (OR ~0.91–0.92), whereas clindamycin, fluoroquinolones and cephalosporins show the largest increases (Figure 1a).
2. **Dedicated meta-analysis (Tariq et al., *Clinical Infectious Diseases*, 2018).** Six studies (4 case-control, 2 cohort; recruitment 1993–2012) show that tetracyclines are associated with a **decreased** risk of CDI relative to other antibiotics: pooled OR **0.62** (95% CI 0.47–0.81; $p < 0.001$); in the doxycycline subgroup OR **0.55** (95% CI 0.40–0.75). Heterogeneity is substantial ($I^2 = 53\%$) (Figure 1b).
3. **Larger recent dataset (VA, 2024).** In patients with community-acquired pneumonia, doxycycline versus azithromycin reduced CDI risk by ~17% (and by ~45% in patients with a prior history of CDI).

Biological plausibility

Tetracyclines disrupt the anaerobic colonic microbiota less than clindamycin, cephalosporins and fluoroquinolones; in addition, doxycycline and tigecycline can reduce toxin production and possess anti-inflammatory properties independent of antibacterial activity.

Limitations (fundamental)

- **All human evidence is observational** — no RCT. Confounding by indication is the principal threat.

- “Lower risk,” not “protective therapy”: the evidence supports **preferring** a tetracycline when clinically appropriate (an antimicrobial-stewardship argument), not prophylactic or therapeutic use against CDI.
- Non-trivial heterogeneity ($I^2 \sim 53\%$).
- Tetracycline resistance is common in clinical *C. difficile* isolates (often via the mobile *tetM* element) — an ecological observation, distinct from patient-level risk.



Figure 1

Figure 1. (a) Community-associated CDI risk by antibiotic class (odds ratio vs no exposure; tetracyclines, in red, are the only class compatible with no increase in risk). (b) Pooled effect of tetracyclines vs other antibiotics (Tariq et al. 2018). Log scale; the dashed line marks OR = 1.

Part 2 — Thesis proposal: a Bayesian meta-analysis

Proposed title

“Tetracyclines and the risk of *Clostridioides difficile* infection: a Bayesian random-effects meta-analysis with prior-sensitivity analysis”

Methodological rationale

The existing frequentist meta-analysis (Tariq 2018) has two features that make a Bayesian approach particularly suitable: - **Few estimates ($k = 6$) and marked heterogeneity ($I^2 \approx 53\%$):** with few studies, the frequentist estimate of the between-study variance (τ^2) is unstable and confidence intervals tend to be too narrow. A Bayesian hierarchical model estimates τ^2 with its full uncertainty. - **Observational nature with potential confounding:** priors allow scepticism toward observational estimates to be incorporated explicitly, and let us quantify how much the conclusions depend on that scepticism.

Objectives

1. **Primary:** estimate the pooled OR (with a 95% credible interval) of the tetracycline–CDI association using a Bayesian hierarchical random-effects model.
2. **Secondary:** (a) estimate τ^2 and the predictive distribution for a new study; (b) compute the **posterior probability** that OR < 1 and that OR < 0.80 (a clinically relevant threshold); (c) doxycycline subgroup analysis; (d) prior-sensitivity analysis.

Question and design (PECO)

- **P** — patients exposed to systemic antibiotics;
- **E** — exposure to a tetracycline (doxycycline, minocycline, tigecycline);
- **C** — exposure to other antibiotics / no tetracycline;
- **O** — incident CDI (microbiological/clinical diagnosis).
- **Design:** systematic review + Bayesian meta-analysis, conducted per **PRISMA 2020**;

protocol pre-registered on **PROSPERO**.

Methods (work plan for the student)

1. Search and selection. Search strings on PubMed/MEDLINE, Embase, Web of Science and Cochrane, from 2016 (end of Tariq's search) to the present, to update the corpus. Two independent reviewers, conflict resolution, PRISMA flow diagram. Quality assessed with the **Newcastle-Ottawa Scale**.

2. Data extraction. For each study: design, setting, population, sample size, CDI definition, adjusted OR/HR and CI, adjustment covariates. Convert all estimates to the **log(OR)** scale with their standard error.

3. Statistical model (random-effects, hierarchical). For study i , with observed estimate $y_i = \log(\text{OR}_i)$ and standard error s_i :

$y_i \sim \text{Normal}(\theta_i, s_i^2)$	(likelihood, s_i known)
$\theta_i \sim \text{Normal}(\mu, \tau^2)$	(between-study random effects)
$\mu \sim \text{Normal}(0, 1^2)$	(prior on the pooled effect – see below)
$\tau \sim \text{Half-Normal}(0, 0.5)$	(weakly informative prior on heterogeneity)

- μ is the pooled effect on the log(OR) scale; report $\exp(\mu)$ as the OR.
- τ is the between-study standard deviation; a *half-Normal* or *half-Cauchy* is preferable to a uniform prior when k is small.

4. Prior-sensitivity analysis (the methodological core of the thesis). Compare at least three priors on μ : - **non-informative / vague:** $\text{Normal}(0, 10^2)$; - **sceptical:** $\text{Normal}(0, 0.35^2)$ — centres expectation on no effect, penalises large effects; - **enthusiastic / literature-informed:** $\text{Normal}(\log 0.62, \sim 0.10^2)$, centred on Tariq's estimate. Show how the posterior estimate and $P(\text{OR} < 1)$ vary across priors (robustness).

5. Computation. MCMC (Hamiltonian Monte Carlo). Recommended tools, all free: - **R** with `brms` or `rjags/R2jags`, or the `bayesmeta` package (designed exactly for this model and very easy to use for a thesis); - alternatively **Python** with `PyMC`. Convergence diagnostics: $\hat{R} < 1.01$, adequate ESS, trace plots.

6. Output and presentation. - Bayesian forest plot with credible intervals and the pooled estimate (like Figure 1b); - posterior density of μ (like Figure 2a) and the estimate of τ ; - a table of $P(\text{OR} < 1)$ and $P(\text{OR} < 0.80)$ for each prior; - **predictive distribution** for a new study; - numerical comparison with Tariq's frequentist estimate.



Figure 2

Figure 2. Key concepts of the method. (a) Bayesian updating: the sceptical *prior* (blue) combined with the data likelihood (grey) yields the posterior distribution (red) of the pooled effect. (b) Hierarchical model: individual study estimates (grey) are “shrunk” toward the common effect (red line) in inverse proportion to their precision. **The study-level values in panel (b) are illustrative**, for teaching purposes.

Why it makes a good thesis

- **Feasible** within a few months: small dataset ($k \approx 6-10$ studies), a standard model, free and well-documented software.
- **Rigorous and instructive:** teaches PRISMA systematic review, Bayesian inference, MCMC, and communication of uncertainty.
- **Clinically relevant:** translates the evidence into a directly interpretable probability (“what is the probability that tetracyclines truly reduce CDI risk?”), useful for stewardship.
- **Original relative to the literature:** no published Bayesian synthesis exists on this specific question; updating the post-2016 corpus adds value.

Risks and mitigations

- *Few new studies since 2016* → the thesis remains valid as a **Bayesian re-analysis** of the existing corpus plus any update.
- *Heterogeneity in exposure/outcome definitions* → subgroup analysis and meta-regression (if k allows) or qualitative discussion.
- *Residual confounding (observational data)* → stated explicitly; the sceptical prior also serves this purpose.

Indicative timeline (6 months)

Month	Activity
1	Protocol, PROSPERO registration, search strings
2	Screening, selection, quality assessment (2 reviewers)
3	Data extraction, construction of the log(OR) dataset
4	Bayesian model implementation, MCMC diagnostics
5	Prior-sensitivity analysis, figures
6	Writing, discussion, final revision

Clinical note: this document is for research and educational purposes. Treatment decisions for individual patients rest with the treating physician, who has access to the full clinical picture.